

	INDIAN INSTITUTE OF TECHNOLOGY ROORKEE
Cloud Security (CST-109)	

Assignment 6

80 marks

(Network Security)

Task 1: Network Access Control and Least Privilege Implementation

If you already have implemented a 3-tier architecture use it to do this task. Other wise implement a 3-tier architecture with following configuration:

Client → Web Server → (Port 80/443)
 Web Server → Database → (Port 3306 for MySQL)
 No direct Client → Database access

The following are some of the required and unnecessary access:

Component	Required Access	Must Be Blocked
Client	HTTP/HTTPS to Web Server	Direct DB access
Web Server	DB access (specific port)	SSH from public
Database	Only Web Server access	All external traffic

- Implement the firewall rules for the above on WebServer, Database server and on client side.
- Restrict the inter-service communication. For example: (a) Web server cannot access unnecessary services (e.g., FTP, SMTP). (b) Database cannot initiate outbound connections
- Test and analyze how excessive permissions increase security risks and how least privilege reduces potential attack vectors.

Task 2: Intrusion Detection System Implementation

- Install and configure an intrusion detection system such as **Snort** on a Linux-based environment. After successful installation, create and customize detection rules to identify at least three types of malicious activities, including **port scanning, ICMP flooding, and unauthorized access attempts**.
- Execute controlled attack scenarios to trigger these rules and observe the generated alerts.
- Submission List: Students are required to describe the custom rules they have written, and provide evidence of detected attacks through logs or screenshots, along with a brief explanation what technique IDS used to identify such threats.

Task 3: Log Analysis and SIEM Integration

Using the logs generated from the configured IDS, perform log collection and analysis by integrating them with a monitoring and visualization platform such as ELK Stack. You are required to process and visualize the collected data to identify patterns such as **frequency of attacks, distribution of source IP addresses, and types of detected threats**. The task involves setting up a basic pipeline for log ingestion and creating visual dashboards for analysis.

Task 4: Network Traffic Analysis using Wireshark

- A) Start packet capture on your active network interface using Wireshark and record traffic for less than 5 minutes while performing activities such as web browsing, searching, and accessing different websites.
- B) Identify source and destination IP addresses and corresponding port numbers for at least three different communication sessions. Give the details of the protocols and the ports and endpoints.
- C) Locate and analyze at least one complete TCP three-way handshake. Explain each step (SYN, SYN-ACK, ACK) and how a connection is established.
- D) Filter DNS traffic and identify at least three domain name queries made by your system. Analyze the corresponding responses and explain how domain resolution works.
- E) Examine whether any data is transmitted in plaintext (e.g., HTTP). Compare it with encrypted traffic (HTTPS/TLS) and explain the security implications.